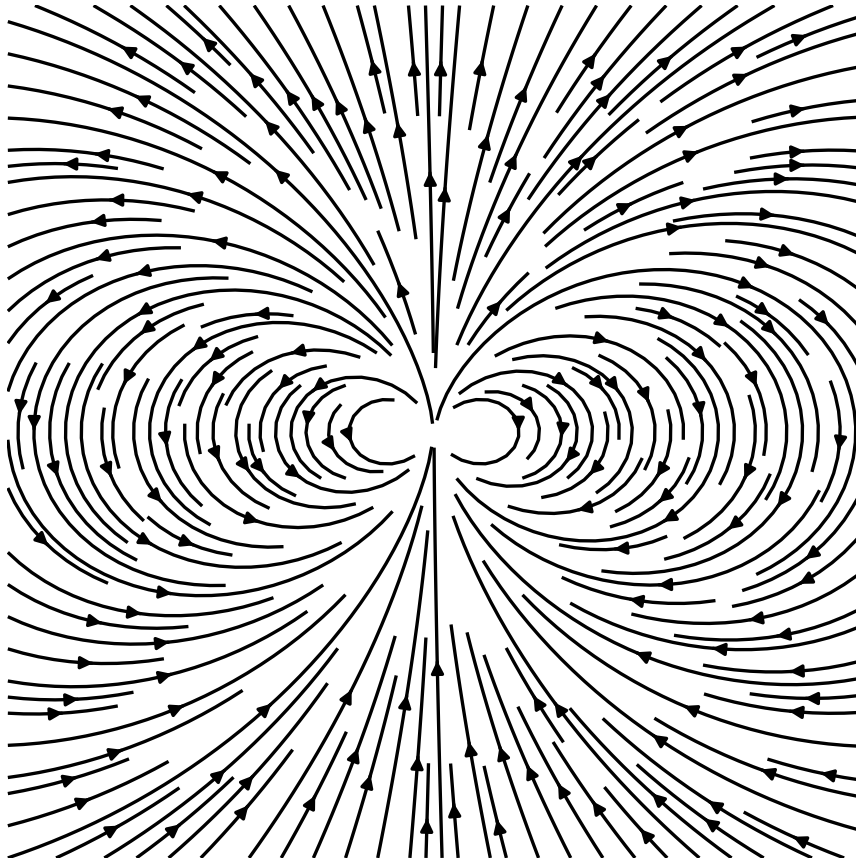




Lecture Notes on Electromagnetic Theory

PHYSICS 131

VERCIL JUAN
IV - BS Physics

1st Semester, AY 2026-2027

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}$$

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{j} + \frac{1}{c^2} \frac{\partial \mathbf{E}}{\partial t}$$

Physics 131 - Electromagnetic Theory I

Course Information

Course Description

Description: Electrostatics in a vacuum, electrostatics in dielectric media, boundary value methods in electrostatics, electric currents, conducting media, magnetostatics in a vacuum, macroscopic and microscopic magnetism, Faraday's law of electromagnetic induction.

Credits: 3.0

Instructor Details

Instructor: Dr. Francis N. C. Paraan

E-mail: `fparaan.upd.edu.ph`

Lecture Hours: WF, 8:30 - 10:00 AM

Course Requirements

You must submit two (2) notebook files, one (1) problem set, and take one sit-in midterm exam to complete the course. Notebooks and problem sets may be handwritten or typeset, but the submission must be a PDF file to be submitted through an online form. The submission link will be provided on the course webpage. You must submit your work by the deadline, otherwise the work will be considered a non-submission. The submission bin will close 48 hours after the deadline as a grace period, and late submissions within this period must be accompanied by a University-recognized excuse.

On-site attendance will be checked to comply with University attendance requirements.

Progress will be monitored by performing two types of checkpoint activities:

1. Notebook Entries
2. Problem Sets

Main References

The Syllabus for the course can be found here:

[COURSE SYLLABUS](#)

The main reference for this course is:

[Introduction to Electrodynamics](#) 4th Edition, David J. Griffiths (2013)

Reminders

Exam Dates

☐ Midterm Exam :

Problem Set Dates

☐

Notebook Submission Dates

☐

☐

Contents

Course Information	2
Main References	3
Reminders	4
Unit 1: Electostatics	8
1.1 Review of Vector Calculus	8
1.1.1 Vectors and Scalars	8
1.1.1.1 Vector operations	9
1.1.2 Vector Properties	10
1.1.2.1 The Vector Norm	10
1.1.2.2 Vector Axioms	10
1.1.2.3 Vector derivatives and Integrals	11
1.1.3 Field	12
1.1.4 The Del Operator	12
1.1.5 Gradient	13
1.1.6 Divergence	14
1.1.7 Curl	14
1.1.8 Laplacian	15
1.1.9 Operations on these Operators	16
1.1.9.1 Product Rule	16
1.1.9.2 Chain Rule	17
1.1.9.3 Quotient Rules	17
1.1.9.4 Additional Rules	17
1.1.10 Line, Surface, and Volume Integrals	17
1.1.10.1 Line integrals	17
1.1.10.2 Surface Integrals	19
1.1.10.3 Volume Integrals	20
1.1.11 Gauss and Stokes Theorem	20
1.1.11.1 Gauss's Divergence Theorem	20
1.1.11.2 Stokes' Theorem	21
1.1.12 Dirac Delta Point Source	21
1.1.13 Potential fields	22

1.1.13.1	Irrotational fields	22
1.1.13.2	Solenoidal Fields	23
1.1.14	Laplacian in orthogonal curvilinear coordinates	24
1.1.14.1	Verification: Cylindrical coordinates	24
1.1.14.2	Verification: Spherical coordinates	25
1.1.15	Divergence of a radial inverse-square field and the Dirac delta	25
1.2	The Electric Field	27
1.3	Boundary Conditions, work and energy, conductors	28
1.4	Calculating Scalar Potentials I	29
1.5	Calculating Scalar Potentials II	30
1.6	Dielectrics	31
1.7	Electric Displacement	32
1.8	Energy in dielectrics, Magnetostatics	33
1.9	Magnetic Vector Potential	34
1.10	Unit Summary	35
Unit 2: Magnetostatics		36

List of Figures

1	Types of Fields	12
2	A scalar field and its gradient	13
3	Examples of vector fields with positive and negative divergences	14
4	Example of vector fields with positive and negative curl	15
5	Scalar field and its Laplacian	16

Unit 1

Electostatics

1.1 Review of Vector Calculus

This is basically just a summary of the **Physics 116 curriculum**, and as such, **I will be pasting an abridged copy of my notes there to here.** Anyway.

1.1.1 Vectors and Scalars

A **scalar** $a \in \mathbb{R}$ is a quantity with magnitude. For the purposes of this notes, scalars are denoted in italic script. A scalar is in the set of real numbers

$$a \in \mathbb{R} \quad (1.1.1)$$

An example of a scalar is the temperature in a room, T_0 , time t , and other single valued quantities.

A **vector** is a collection of scalars, usually indicating magnitude and direction. We represent vectors in math bold face. A vector of n dimensions is represented by n numbers, v_i , where i represents the i -th element of the vector. A vector can be a column vector or a row vector.

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \quad \mathbf{v}^\top = \begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix} \quad (1.1.2)$$

An example of a vector is a particle's position in space

$$\mathbf{r} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad (1.1.3)$$

A tensor is a generalization of vectors. It is essentially an n dimensional (More commonly called rank) vector. This is generally beyond the scope of this notes, but I included it for generality. An example of a tensor is the stress energy tensor.

$$T = \begin{bmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{bmatrix} \quad (1.1.4)$$

It's essentially a matrix, but more general. A matrix is a rank 2 tensor. A vector is a rank 1 tensor, while a scalar is a rank 0 tensor.

1.1.1.1 Vector operations

There are several operations that can be done with vectors and scalars. Here are some of them.

Vector addition For a vector $\mathbf{u}$ and $\mathbf{v}$, adding them is done by adding element-wise. The same is true for subtraction.

$$\mathbf{u} \pm \mathbf{v} = (u_i \pm v_i)\hat{\mathbf{e}}_i \quad (1.1.5)$$

Scalar multiplication A vector multiplied by a scalar is equivalent to multiplying all its elements with the scalar. As a shorthand, we can represent vectors and all its elements with one expression $\mathbf{v} = v_i$, where the index i runs over from 1 to n . Given $a \in \mathbb{R}$, a scalar multiplication is

$$a\mathbf{v} = \begin{bmatrix} av_1 \\ av_2 \\ av_3 \end{bmatrix} = av_i \quad (1.1.6)$$

Dot Product The dot product of two vectors is the sum of the product of the i -th element of each vector. For example, given a vector $\mathbf{v} = v_i$ and $\mathbf{u} = u_i$, their dot product is

$$\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2 + u_3v_3 = \sum_i u_iv_i \quad (1.1.7)$$

We will encounter a lot of these sums in general, so we introduce the *Einstein summation convention* where, if we see repeated indices, this implies that the expression is being summed over that index. As such, we can more conveniently define a dot product as

$$\mathbf{u} \cdot \mathbf{v} = u_iv_i \quad (1.1.8)$$

Cross Product The cross product is pretty hard to define, but in general, it is shown the vector which is the determinant of the matrix of the vector elements. But first, we introduce the following vectors

$$\hat{\mathbf{e}}_1 = \hat{\mathbf{i}} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \hat{\mathbf{e}}_2 = \hat{\mathbf{j}} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \hat{\mathbf{e}}_3 = \hat{\mathbf{k}} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad (1.1.9)$$

There are known as basis vectors of the vector space. Basically unit vectors of a single dimension. We can represent any vector in $\mathbb{R}^3$ in this notation.

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = v_1\hat{\mathbf{i}} + v_2\hat{\mathbf{j}} + v_3\hat{\mathbf{k}} = v_i\hat{\mathbf{e}}_i \quad (1.1.10)$$

It must be noted that $\mathbf{v} = v_i(x, y, z)\hat{\mathbf{e}}_i$. In that definition, then, we often see the dot product defined as

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = (u_2v_3 - u_3v_2)\hat{\mathbf{i}} - (u_1v_3 - u_3v_1)\hat{\mathbf{j}} + (u_1v_2 - u_2v_1)\hat{\mathbf{k}} \quad (1.1.11)$$

The Kronecker Delta and LeviCivita operators The Kronecker delta is an operator which basically turns on or of a product depending on it's indices. Formally, it is defined as

$$\delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \quad (1.1.12)$$

and the Levi-Civita symbol is defined to be 1 if the indices are an even permutation of i, j, k and zero if it is not.

$$\epsilon_{ijk} = \begin{cases} 1 & \text{if } ijk \text{ is } 123, 312, 231 \\ -1 & \text{if } ijk \text{ is } 132, 321, 213 \\ 0 & \text{if } i = j \text{ or } j = k \text{ or } i = k \end{cases} \quad (1.1.13)$$

with this, we can more succintly define the dot and cross product as

$$\mathbf{u} \cdot \mathbf{v} = \delta_{ij} u_i v_j \quad (1.1.14)$$

$$\mathbf{u} \times \mathbf{v} = \epsilon_{ijk} u_i v_j \hat{\mathbf{e}}_k \quad (1.1.15)$$

Einstein Summation Notation To prevent clunkiness of equations, we oftem omit the summation symbol. By convention, when an index appears **twice** in a single term (once as a superscript and once as a subscript), it is **implicitly summed over** all values of that index.

$$\sum_i a_i b^i \equiv a_i b^i \quad (1.1.16)$$

The summation symbol $\sum$ is ommited. This applies to the tensor notation listend above. Sometimes, it is also not done with superscripts, but the summation convention applies nonetheless.

$$\sum_i a_i b_i \equiv a_i b_i \quad (1.1.17)$$

1.1.2 Vector Properties

1.1.2.1 The Vector Norm

The vector norm is the *length* of the vector, so to speak. It is defined as

$$\|\mathbf{v}\| = \sqrt{\mathbf{v}^2} = \sqrt{\mathbf{v} \cdot \mathbf{v}} \quad (1.1.18)$$

Note that we define the square of a vector to be its dot product with itself. $\mathbf{v}^2 = \mathbf{v} \cdot \mathbf{v}$

1.1.2.2 Vector Axioms

Technically, a vector is really just defined as an element of a vector space, which is an abstract space which obeys the following properties¹.

¹This comes from the Math 40 curriculum

Definition 1.1 : Vector Axioms

Let $\mathbf{u}, \mathbf{v}$ and $\mathbf{w}$ be vectors in $\mathbb{V}$. The space $\mathbb{V}$ is a vector space if the following conditions hold:

These operations are defined as follows:

1. Vector addition, $+: V \times \mathbb{V} \rightarrow \mathbb{V}$
2. Scalar multiplication, $\cdot: \mathbb{R} \times \mathbb{V} \rightarrow \mathbb{V}$

and these conditions must hold

1. Closure under addition, $\mathbf{u} + \mathbf{v} \in \mathbb{V}$
2. Commutativity, $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
3. Associativity, $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{v} + (\mathbf{u} + \mathbf{w})$
4. Additive identity (zero vector $\hat{0}$), $\forall \mathbf{v} \in \mathbb{V}, \exists \hat{0} \in \mathbb{V} : \mathbf{v} + \hat{0} = \mathbf{v}$
5. Additive inverse, $\forall \mathbf{v} \in \mathbb{V}, \exists -\mathbf{v} \in \mathbb{V} : \mathbf{v} + (-\mathbf{v}) = \hat{0}$
6. Closure under scalar multiplication, $\forall a \in \mathbb{R} \wedge \mathbf{v} \in \mathbb{V}, a\mathbf{v} \in \mathbb{V}$
7. Distributivity of scalar multiplication under vector addition, $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$
8. Distributivity of scalar multiplication under scalar addition, $\forall a, b \in \mathbb{R}, (a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$
9. Associativity of scalar multiplication, $a(b\mathbf{v}) = (ab)\mathbf{v}$
10. Existence of multiplicative identity, $\forall \mathbf{v} \in \mathbb{V}, \exists 1 : 1 \cdot \mathbf{v} = \mathbf{v}$

1.1.2.3 Vector derivatives and Integrals

To take the derivative of a vector, one must apply the derivative on all of its components, just like normal derivatives. Chain rule and other calculus methods still apply.

$$\frac{d\mathbf{v}}{dt} = \begin{bmatrix} \frac{dv_1}{dt} \\ \frac{dv_2}{dt} \\ \frac{dv_3}{dt} \end{bmatrix} = \frac{dv_i}{dt} \quad (1.1.19)$$

Similarly, the same applies for integration, along with path integrals and the like which will not be part of this section.

$$\int \mathbf{v} dt = \begin{bmatrix} \int v_1 dt \\ \int v_2 dt \\ \int v_3 dt \end{bmatrix} = \int v_i dt \quad (1.1.20)$$

1.1.3 Field

A *field* is a function that assigns a value to every point in a defined space. It can be:

- **Scalar field:** assigns a scalar value, e.g., temperature $T(x, y, z)$. This returns a space where every point has a scalar value.
- **Vector field:** assigns a vector value, e.g., velocity $\mathbf{v}(x, y, z)$. This returns a space where every point has a vector value.

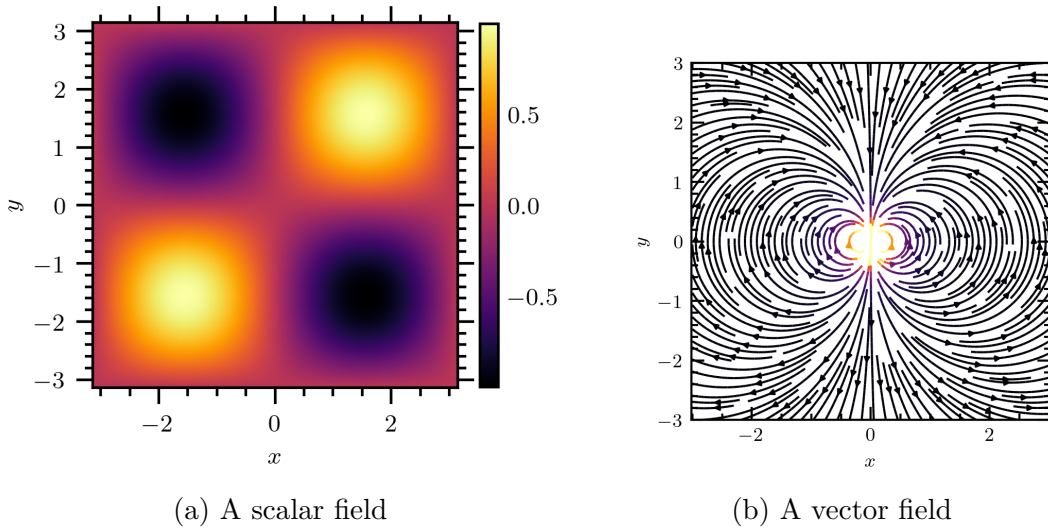


Figure 1: Types of Fields

1.1.4 The Del Operator

The Del operator ∇ is the generalization of the derivative. We define it as a vector of the partial derivative with respect to the dimension, i.e.

Definition 1.2 : ∇ operator

$$\nabla = \frac{\partial}{\partial x} \hat{\mathbf{i}} + \frac{\partial}{\partial y} \hat{\mathbf{j}} + \frac{\partial}{\partial z} \hat{\mathbf{k}} = \begin{bmatrix} \partial/\partial x \\ \partial/\partial y \\ \partial/\partial z \end{bmatrix} \quad (1.1.21)$$

As a shorthand, we can represent it as $\frac{\partial}{\partial x_i}$, where x_i is the general dimension. This can be further shorthanded to ∂_i . Thus, a compact vector definition of the Del operator is

$$\nabla = \partial_i \quad (1.1.22)$$

1.1.5 Gradient

We often represent fields as $\psi(x, y, z)$; however, we can just make x, y, z into a vector $\mathbf{r} = (x, y, z)$. In that case, the physical interpretation of the gradient becomes clearer.

The **gradient** of a scalar field $\phi(\mathbf{r})$ is a vector field that points in the direction of the greatest rate of increase of ϕ :

Definition 1.3 : Gradient

$$\nabla\phi = \frac{\partial\phi}{\partial x}\hat{\mathbf{i}} + \frac{\partial\phi}{\partial y}\hat{\mathbf{j}} + \frac{\partial\phi}{\partial z}\hat{\mathbf{k}} = \begin{bmatrix} \frac{\partial\phi}{\partial x} \\ \frac{\partial\phi}{\partial y} \\ \frac{\partial\phi}{\partial z} \end{bmatrix} \quad (1.1.23)$$

It measures how rapidly the scalar field changes and in which direction this change is maximum. For example, in a temperature field $T(x, y, z)$, the gradient ∇T points in the direction where temperature increases most rapidly, and its magnitude gives the rate of increase per unit distance.

In index notation, we can write

$$\nabla\phi = \partial_i\phi, \quad (1.1.24)$$

where ∂_i denotes differentiation with respect to x_i .

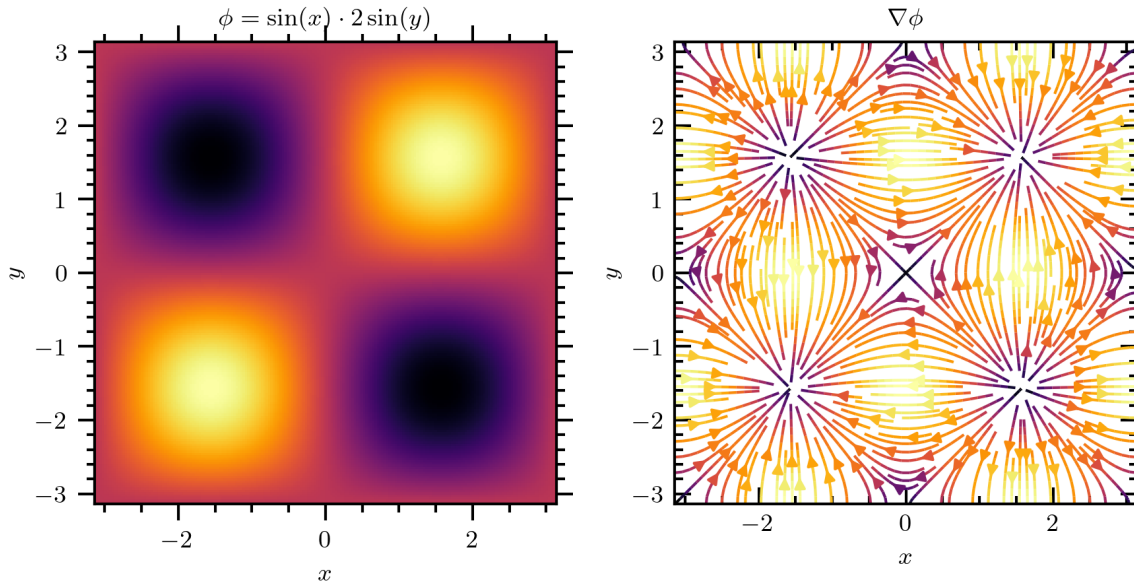


Figure 2: A scalar field and its gradient

1.1.6 Divergence

The **divergence** of a vector field $\mathbf{F}(\mathbf{r}) = [F_x \ F_y \ F_z]^\top$ measures the net flux per unit volume leaving a point:

Definition 1.4 : Divergence

$$\nabla \cdot \mathbf{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}. \quad (1.1.25)$$

A positive divergence indicates a *source* (more field lines leaving than entering), while a negative divergence indicates a *sink* (field lines converging). If $\nabla \cdot \mathbf{F} = 0$, the field is said to be **solenoidal**, meaning it has no net flux out of any closed surface.

This operation is essentially the dot product of the del operator with the vector field:

$$\nabla \cdot \mathbf{F} = \delta_{ij} \partial_i F_j, \quad (1.1.26)$$

Divergence appears naturally in continuity equations. For instance, $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$ in Gauss's law for electricity, where it represents the electric charge density acting as a source of the field.

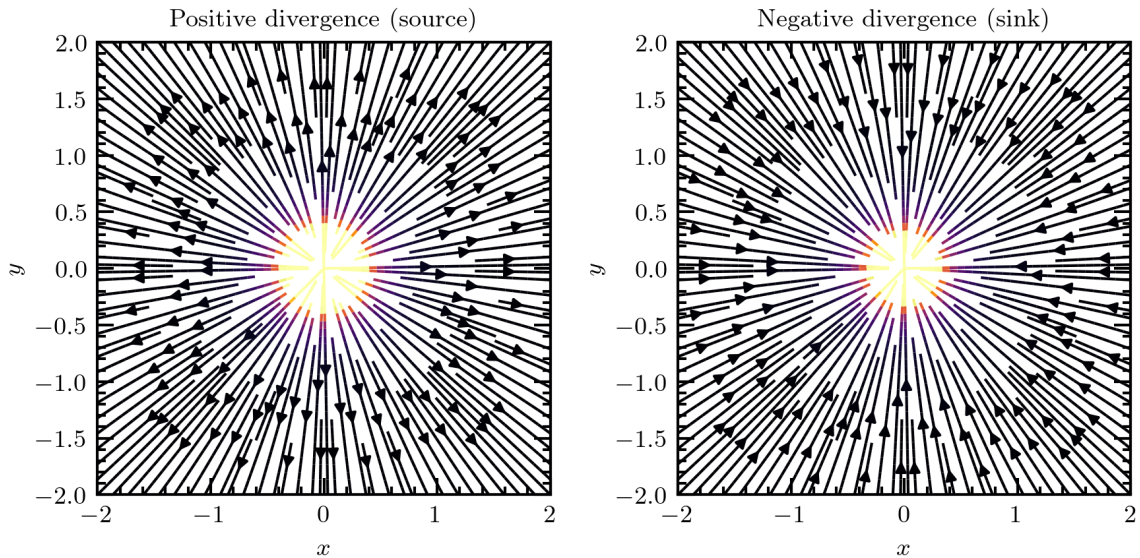


Figure 3: Examples of vector fields with positive and negative divergences

1.1.7 Curl

The **curl** of a vector field $\mathbf{F}(\mathbf{r})$ measures its local rotation or the tendency to induce circulation around a point:

Definition 1.5 : Curl

$$\nabla \times \mathbf{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix}. \quad (1.1.27)$$

The direction of $\nabla \times \mathbf{F}$ gives the axis of rotation (using the right-hand rule), and its magnitude gives the rotational strength.

This is equivalent to taking the cross product of the del operator with the vector field:

$$\nabla \times \mathbf{F} = \varepsilon_{ijk} \partial_i F_j \hat{\mathbf{e}}_k \quad (1.1.28)$$

In physical contexts, the curl represents quantities such as the rotational velocity of a fluid or the magnetic field induced by a current. For example, in electromagnetism, $\nabla \times \mathbf{E} = -\partial \mathbf{B} / \partial t$ expresses Faraday's law of induction.

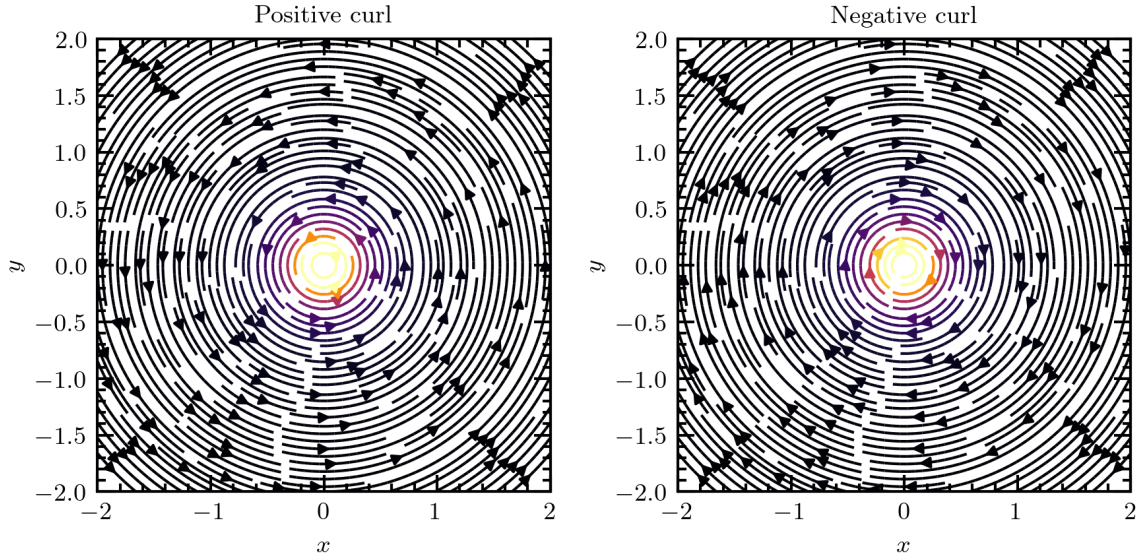


Figure 4: Example of vector fields with positive and negative curl

1.1.8 Laplacian

The **Laplacian** operator acts on scalar or vector fields and is defined as the divergence of the gradient:

Definition 1.6 : Laplacian

$$\nabla^2 \phi = \nabla \cdot (\nabla \phi) = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2}. \quad (1.1.29)$$

It measures how a field value at a point compares to its average in the surrounding region — in other words, how “curved” or “peaked” the field is locally. It is worthy to note that the laplacian of a scalar field is also a scalar field.

In index notation, this becomes

$$\nabla^2 \phi = \delta_{ij} \partial_i \partial_j \phi. \quad (1.1.30)$$

The Laplacian appears in many fundamental physical equations:

- Poisson's equation: $\nabla^2 \phi = -\rho/\varepsilon_0$ (electrostatics)
- Heat equation: $\partial \phi / \partial t = \alpha \nabla^2 \phi$ (diffusion)
- Schrödinger equation: $\nabla^2 \psi$ governs spatial variations in quantum wavefunctions
- Newtonian Field equation for Gravity: $\nabla^2 \Phi = 4\pi G \rho$

If $\nabla^2 \phi = 0$, the field is called **harmonic**, meaning it satisfies Laplace's equation and represents an equilibrium distribution such as steady-state temperature or potential.

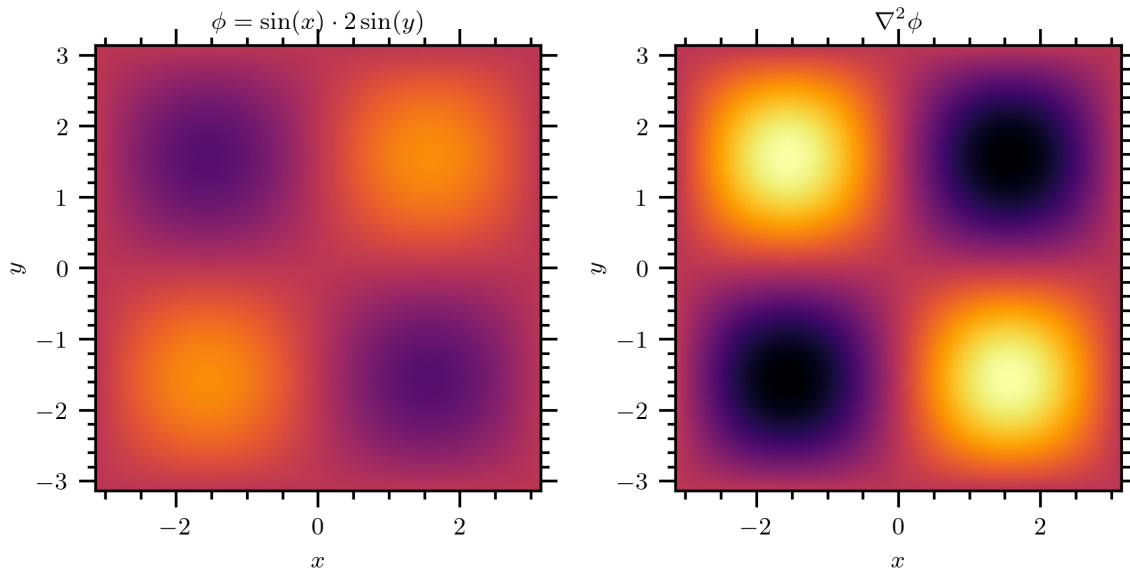


Figure 5: Scalar field and its Laplacian

1.1.9 Operations on these Operators

The usual differentiation rules still apply for ∇ , $\nabla \cdot$, and $\nabla \times$, because they are built from ordinary partial derivatives. One only has to account for the fact that the fields may be scalar or vector valued.

1.1.9.1 Product Rule

For scalar fields f, g

$$\nabla(fg) = f \nabla g + g \nabla f \quad (1.1.31)$$

For a scalar f and a vector field $\mathbf{A}$

$$\nabla \cdot (f\mathbf{A}) = f(\nabla \cdot \mathbf{A}) + (\nabla f) \cdot \mathbf{A} \quad (1.1.32)$$

and

$$\nabla \times (f\mathbf{A}) = f(\nabla \times \mathbf{A}) + (\nabla f) \times \mathbf{A} \quad (1.1.33)$$

1.1.9.2 Chain Rule

This also applies for these operations.

Suppose $f = f(u)$ and $u = u(x_1, x_2, x_3)$, then

$$\nabla f(u) = \frac{df}{du} \nabla u \quad (1.1.34)$$

More generally, if $f = f(u, v)$ and $u = u(x, y, z)$ and $v = v(x, y, z)$

$$\nabla f(u, v) = \frac{\partial f}{\partial u} \nabla u + \frac{\partial f}{\partial v} \nabla v \quad (1.1.35)$$

1.1.9.3 Quotient Rules

For scalar fields f, g and vector field $\mathbf{A}$, we have three possible quotient rules.

$$\begin{aligned} \nabla \left(\frac{f}{g} \right) &= \frac{g \nabla f - f \nabla g}{g^2} \\ \nabla \cdot \left(\frac{\mathbf{A}}{g} \right) &= \frac{g(\nabla \cdot \mathbf{A}) - \mathbf{A} \cdot (\nabla g)}{g^2} \\ \nabla \times \left(\frac{\mathbf{A}}{g} \right) &= \frac{g(\nabla \times \mathbf{A}) + \mathbf{A} \times (\nabla g)}{g^2} \end{aligned}$$

1.1.9.4 Additional Rules

Some other useful identities are

$$\begin{aligned} \nabla \cdot (\mathbf{A} + \mathbf{B}) &= \nabla \cdot \mathbf{A} + \nabla \cdot \mathbf{B} \\ \nabla \times (\mathbf{A} + \mathbf{B}) &= \nabla \times \mathbf{A} + \nabla \times \mathbf{B} \end{aligned}$$

And other second derivative identities

$$\begin{aligned} \nabla \cdot (\nabla f) &= \nabla^2 f \\ \nabla \times (\nabla f) &= 0 \\ \nabla \cdot (\nabla \times \mathbf{A}) &= 0 \end{aligned}$$

1.1.10 Line, Surface, and Volume Integrals

1.1.10.1 Line integrals

For reference, a line integral is defined as the integral over the path Γ of a parameterized vector path function $\mathbf{r}(t)$ over a scalar field ϕ or a vector field $\mathbf{F}$ with respect to a parameter t . Go figure. It comes in two forms: scalar line integrals and vector line integrals.

For a scalar field, the line integral is

Definition 1.7 : Scalar line Integral

$$\int_{\Gamma} \phi(\mathbf{r}) \, ds = \int_a^b \phi(\mathbf{r}(t)) \left\| \frac{d\mathbf{r}}{dt} \right\| dt. \quad (1.1.36)$$

Here, ds is the differential arc length along the curve. Since

$$ds = \left\| \frac{d\mathbf{r}}{dt} \right\| dt, \quad (1.1.37)$$

the scalar line integral can be understood as summing the value of the scalar field along the curve, weighted by the infinitesimal distance traveled along that curve. It is therefore independent of the particular parameterization of the curve, provided that the parameterization preserves its orientation and traces the same path.

For a vector field, the line integral is

Definition 1.8 : Vector line Integral

$$\int_{\Gamma} \mathbf{F}(\mathbf{r}) \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \frac{d\mathbf{r}}{dt} dt. \quad (1.1.38)$$

Unlike the scalar line integral, the vector line integral depends on the direction in which the path is traversed. Reversing the orientation of Γ gives

$$\int_{-\Gamma} \mathbf{F} \cdot d\mathbf{r} = - \int_{\Gamma} \mathbf{F} \cdot d\mathbf{r}. \quad (1.1.39)$$

The quantity $d\mathbf{r}$ is the infinitesimal displacement vector along the curve,

$$d\mathbf{r} = \frac{d\mathbf{r}}{dt} dt, \quad (1.1.40)$$

so the vector line integral measures the component of the vector field along the direction of motion. In physics, this is why the work done by a force $\mathbf{F}$ along a path Γ is written as

$$W = \int_{\Gamma} \mathbf{F} \cdot d\mathbf{r}. \quad (1.1.41)$$

If the vector field is conservative, so that

$$\mathbf{F} = -\nabla\phi, \quad (1.1.42)$$

then the integral depends only on the endpoints of the curve rather than on the path taken. In that case,

$$\int_{\Gamma} \mathbf{F} \cdot d\mathbf{r} = \phi(\mathbf{r}_a) - \phi(\mathbf{r}_b), \quad (1.1.43)$$

where $\mathbf{r}_a$ and $\mathbf{r}_b$ are the initial and final points of the path. Equivalently, for a closed curve,

$$\oint_{\Gamma} \mathbf{F} \cdot d\mathbf{r} = 0. \quad (1.1.44)$$

This gives a useful connection between line integrals and the gradient: gradients of scalar fields are conservative vector fields. More generally, in a simply connected region,

$$\nabla \times \mathbf{F} = \mathbf{0} \quad \Longleftrightarrow \quad \mathbf{F} = \nabla \phi \quad (1.1.45)$$

for some scalar potential ϕ , subject to the appropriate regularity conditions.

1.1.10.2 Surface Integrals

A surface integral is the analogous construction for a two-dimensional surface embedded in three-dimensional space. Let a surface Σ be parameterized by two parameters u and v ,

$$\mathbf{r} = \mathbf{r}(u, v), \quad (u, v) \in D. \quad (1.1.46)$$

The tangent vectors to the surface are

$$\frac{\partial \mathbf{r}}{\partial u}, \quad \frac{\partial \mathbf{r}}{\partial v}, \quad (1.1.47)$$

and therefore the infinitesimal area element is

$$dS = \left\| \frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \right\| du dv. \quad (1.1.48)$$

For a scalar field ϕ , the surface integral is

Definition 1.9 : Scalar surface integral

$$\int_{\Sigma} \phi dS = \iint_D \phi(\mathbf{r}(u, v)) \left\| \frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \right\| du dv. \quad (1.1.49)$$

This is the two-dimensional analogue of the scalar line integral: rather than summing the field along a curve, we sum it over an entire surface, weighted by the infinitesimal area element.

For a vector field $\mathbf{F}$, however, the natural surface integral is a *flux integral*. The oriented surface element is defined by

$$d\mathbf{S} = \left(\frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \right) du dv, \quad (1.1.50)$$

where the direction of $d\mathbf{S}$ is determined by the chosen orientation of the surface. The flux of $\mathbf{F}$ through Σ is then

Definition 1.10 : Vector surface integral

$$\int_{\Sigma} \mathbf{F} \cdot d\mathbf{S} = \iint_D \mathbf{F}(\mathbf{r}(u, v)) \cdot \left(\frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \right) du dv. \quad (1.1.51)$$

The dot product selects the component of the vector field normal to the surface. Thus, the flux measures how much of the field passes through the surface rather than how much of it runs along the surface.

1.1.10.3 Volume Integrals

Finally, a volume integral is an integral over a three-dimensional region V . For a scalar field ϕ , we write

$$\int_V \phi dV. \quad (1.1.52)$$

In Cartesian coordinates,

$$dV = dx dy dz, \quad (1.1.53)$$

so that

Definition 1.11 : Volume integral

$$\int_V \phi dV = \iiint_V \phi(x, y, z) dx dy dz. \quad (1.1.54)$$

These constructions form the basis for the integral versions of the fundamental theorems of vector calculus, including Green's theorem, Stokes' theorem, and the divergence theorem. These theorems relate derivatives of a field within some region to the behavior of that field on the boundary of the region.

1.1.11 Gauss and Stokes Theorem

Gauss's and Stokes' theorems are two of the most important results in vector calculus, providing relationships between surface and volume (or line and surface) integrals. They serve as generalizations of the Fundamental Theorem of Calculus to higher dimensions, allowing one to convert integrals over regions into integrals over their boundaries.

1.1.11.1 Gauss's Divergence Theorem

The divergence theorem, also known as Gauss's theorem, relates the flux of a vector field through a closed surface to the divergence of the field inside the volume enclosed by that surface. Mathematically, it is expressed as

Theorem 1.1 : Gauss' Theorem

$$\iiint_V (\nabla \cdot \mathbf{F}) dV = \oiint_S \mathbf{F} \cdot d\mathbf{S}, \quad (1.1.55)$$

where V is the volume enclosed by the closed surface S , $\mathbf{F}$ is a continuously differentiable vector field, and $d\mathbf{S}$ is the outward-pointing area element on S . This theorem is widely used in electromagnetism, fluid mechanics, and gravitational theory.

This is sometimes shortened as

$$\int_{\Omega} \nabla \cdot \mathbf{F} dV = \oint_{\partial\Omega} \mathbf{F} \cdot d\mathbf{S} \quad (1.1.56)$$

1.1.11.2 Stokes' Theorem

Stokes' theorem relates the circulation of a vector field around a closed curve to the curl of the field over the surface bounded by that curve. It can be written as

Theorem 1.2 : Stokes' Theorem

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}, \quad (1.1.57)$$

where C is the positively oriented boundary curve of the surface S . Stokes' theorem generalizes the concept of Green's theorem in the plane to three dimensions.

Interpretation In essence, both theorems express how local differential properties of vector fields (divergence and curl) are related to global integral quantities (flux and circulation). They form the mathematical foundation for many physical laws, such as Gauss's law in electromagnetism and the conservation of mass and momentum in fluid dynamics.

1.1.12 Dirac Delta Point Source

The **Dirac delta function** $\delta(\mathbf{r})$ (or $\delta^{(3)}(\mathbf{r})$ for three dimensions) models an idealized point source.

Definition 1.12 : Dirac Delta function

$$\delta(x) = \begin{cases} 1 & x = 0 \\ 0 & x \neq 0 \end{cases} \quad (1.1.58)$$

It has the property

$$\int_{-\infty}^{+\infty} \delta(x) dx = 1 \quad (1.1.59)$$

Its defining property is the sifting integral:

Definition 1.13 : Dirac Delta Sifting

$$\int_V f(\mathbf{r}) \delta(\mathbf{r} - \mathbf{r}_0) d^3r = f(\mathbf{r}_0), \quad (1.1.60)$$

for any test function f and any volume V containing $\mathbf{r}_0$. The delta is zero for $\mathbf{r} \neq \mathbf{r}_0$, singular at $\mathbf{r} = \mathbf{r}_0$, and normalized so that its integral over all space is 1:

$$\int_{\mathbb{R}^3} \delta(\mathbf{r}) d^3r = 1. \quad (1.1.61)$$

Delta and differential operators The Dirac delta often appears as the source term in Poisson-like equations. Two classic identities (in three dimensions) are:

$$\nabla^2 \left(\frac{1}{r} \right) = -4\pi \delta^{(3)}(\mathbf{r}), \quad (1.1.62)$$

$$\nabla \cdot \left(\frac{\mathbf{r}}{r^3} \right) = 4\pi \delta^{(3)}(\mathbf{r}). \quad (1.1.63)$$

These relations are to be understood in the distributional sense: the left-hand side vanishes for $\mathbf{r} \neq \mathbf{0}$ but the integral over any volume containing the origin produces the stated constant.

Example: point charge in electrostatics For an electrostatic potential of a point charge q at the origin,

$$\phi(\mathbf{r}) = \frac{q}{4\pi\epsilon_0} \frac{1}{r},$$

Poisson's equation $\nabla^2 \phi = -\rho/\epsilon_0$ is satisfied with $\rho(\mathbf{r}) = q \delta^{(3)}(\mathbf{r})$ because of (1.1.62).

1.1.13 Potential fields

1.1.13.1 Irrotational fields

If the curl of a vector field $\mathbf{F}$ vanishes everywhere, then $\mathbf{F}$ can be written as the gradient of a scalar potential field V .

$$\nabla \mathbf{F} = 0 \iff \mathbf{F} = -\nabla V \quad (1.1.64)$$

Theorem 1.3 : Curl-less or irrotational fields

The following conditions are equivalent.

(a) $\nabla \times \mathbf{F} = 0$

(b) $\int_a^b \mathbf{F} \cdot d\mathbf{l}$ is independent of path for any given end points.

(c) $\oint \mathbf{F} \cdot d\mathbf{l} = 0$

(d) $\mathbf{F} = -\nabla V$

1.1.13.2 Solenoidal Fields

If the divergence of a vector field $\mathbf{F}$ vanishes everywhere, then $\mathbf{F}$ can be written as the curl of a vector potential field $\mathbf{A}$.

$$\nabla \cdot \mathbf{F} = 0 \iff \mathbf{F} = \nabla \times \mathbf{A} \quad (1.1.65)$$

Theorem 1.4 : Divergence-less or solenoidal fields

The following conditions are equivalent.

- (a) $\nabla \cdot \mathbf{F} = 0$
- (b) $\int \mathbf{F} \cdot d\mathbf{a}$ is independent of surface for any given boundary line.
- (c) $\oint \mathbf{F} \cdot d\mathbf{a} = 0$
- (d) $\mathbf{F} = \nabla \times \mathbf{A}$

Remark :

A vector field potential is not restricted to these definitions. A vector field $\mathbf{F}$ can be written as the sum of a gradient of a scalar and a curl of a vector field.

$$\mathbf{F} = -\nabla V + \nabla \times \mathbf{A} \quad (1.1.66)$$

1.1.14 Laplacian in orthogonal curvilinear coordinates

We introduce an orthogonal curvilinear coordinate system (q_1, q_2, q_3) with orthonormal coordinate directions $\hat{\mathbf{e}}_1, \hat{\mathbf{e}}_2, \hat{\mathbf{e}}_3$ and scale factors (metric coefficients)

$$h_i(q_1, q_2, q_3) = \left\| \frac{\partial \mathbf{r}}{\partial q_i} \right\|, \quad i = 1, 2, 3.$$

Infinitesimal displacement is

$$d\mathbf{r} = h_1 dq_1 \hat{\mathbf{e}}_1 + h_2 dq_2 \hat{\mathbf{e}}_2 + h_3 dq_3 \hat{\mathbf{e}}_3,$$

and the volume element is

$$dV = h_1 h_2 h_3 dq_1 dq_2 dq_3.$$

The gradient of a scalar ϕ is

$$\nabla \phi = \sum_{i=1}^3 \frac{1}{h_i} \frac{\partial \phi}{\partial q_i} \hat{\mathbf{e}}_i. \quad (1.1.67)$$

A vector field has components F_i in the $\hat{\mathbf{e}}_i$ basis:

$$\mathbf{F} = \sum_{i=1}^3 F_i \hat{\mathbf{e}}_i.$$

The divergence in orthogonal curvilinear coordinates is

$$\nabla \cdot \mathbf{F} = \frac{1}{h_1 h_2 h_3} \sum_{i=1}^3 \frac{\partial}{\partial q_i} \left(\frac{h_1 h_2 h_3}{h_i} F_i \right). \quad (1.1.68)$$

From this, the Laplacian of a scalar ϕ (divergence of the gradient) becomes

$$\nabla^2 \phi = \frac{1}{h_1 h_2 h_3} \sum_{i=1}^3 \frac{\partial}{\partial q_i} \left(\frac{h_1 h_2 h_3}{h_i^2} \frac{\partial \phi}{\partial q_i} \right) \quad (1.1.69)$$

which is the standard formula for the Laplacian in an orthogonal coordinate system.

1.1.14.1 Verification: Cylindrical coordinates

Cylindrical coordinates (r, θ, z) :

$$\mathbf{r} = (r \cos \theta, r \sin \theta, z), \quad h_r = 1, \quad h_\theta = r, \quad h_z = 1.$$

Compute $h_1 h_2 h_3 = r$. Plugging into (1.1.69):

$$\begin{aligned} \nabla^2 \phi &= \frac{1}{r} \left\{ \frac{\partial}{\partial r} \left(r \frac{\partial \phi}{\partial r} \right) + \frac{\partial}{\partial \theta} \left(\frac{1}{r} \frac{\partial \phi}{\partial \theta} \right) + \frac{\partial}{\partial z} \left(r \frac{\partial \phi}{\partial z} \right) \right\} \frac{1}{1} \\ &= \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \phi}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 \phi}{\partial \theta^2} + \frac{\partial^2 \phi}{\partial z^2}, \end{aligned}$$

which is the familiar cylindrical Laplacian.

1.1.14.2 Verification: Spherical coordinates

Spherical coordinates (r, θ, φ) with θ the polar angle and φ the azimuth:

$$\mathbf{r} = (r \sin \theta \cos \varphi, r \sin \theta \sin \varphi, r \cos \theta), \quad h_r = 1, \quad h_\theta = r, \quad h_\varphi = r \sin \theta.$$

Here $h_1 h_2 h_3 = r^2 \sin \theta$. Plugging into (1.1.69):

$$\begin{aligned} \nabla^2 \phi &= \frac{1}{r^2 \sin \theta} \left\{ \frac{\partial}{\partial r} \left(r^2 \sin \theta \frac{\partial \phi}{\partial r} \right) + \frac{\partial}{\partial \theta} \left(\sin \theta \frac{1}{r^2} r^2 \frac{\partial \phi}{\partial \theta} \right) \right. \\ &\quad \left. + \frac{\partial}{\partial \varphi} \left(\frac{r^2 \sin \theta}{(r \sin \theta)^2} \frac{\partial \phi}{\partial \varphi} \right) \right\} \\ &= \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \phi}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \phi}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 \phi}{\partial \varphi^2}, \end{aligned}$$

the standard spherical Laplacian.

1.1.15 Divergence of a radial inverse-square field and the Dirac delta

Consider the radial inverse-square field in $\mathbb{R}^3$,

$$\mathbf{F}(\mathbf{r}) = k \frac{\hat{\mathbf{r}}}{r^2} = k \frac{\mathbf{r}}{r^3}, \quad r = \|\mathbf{r}\|, \quad \hat{\mathbf{r}} = \mathbf{r}/r,$$

with constant k .

1. Away from the origin For $r \neq 0$ one may compute the divergence using spherical components or directly note that $\mathbf{F}$ is the gradient of a $1/r$ -type potential up to sign:

$$\mathbf{F} = k \frac{\mathbf{r}}{r^3} = -k \nabla \left(\frac{1}{r} \right).$$

Because $\nabla^2(1/r) = 0$ for $r \neq 0$, it follows that

$$\nabla \cdot \mathbf{F} = -k \nabla \cdot \nabla \left(\frac{1}{r} \right) = -k \nabla^2 \left(\frac{1}{r} \right) = 0 \quad (r \neq 0).$$

2. Flux through a sphere and the distributional divergence Let S_R be the sphere of radius R centered at the origin. The outward flux is

$$\Phi(R) = \iint_{S_R} \mathbf{F} \cdot d\mathbf{S} = \iint_{S_R} k \frac{\hat{\mathbf{r}}}{r^2} \cdot (\hat{\mathbf{r}} R^2 d\Omega) = k \iint_{S_R} d\Omega = 4\pi k.$$

This flux is independent of R (for any $R > 0$). By the divergence theorem, for any volume V containing the origin,

$$\int_V \nabla \cdot \mathbf{F} d^3r = \iint_{\partial V} \mathbf{F} \cdot d\mathbf{S} = 4\pi k.$$

If V does *not* contain the origin the flux is zero. These facts characterize the divergence of $\mathbf{F}$ as a distribution:

$$\nabla \cdot \left(k \frac{\mathbf{r}}{r^3} \right) = 4\pi k \delta^{(3)}(\mathbf{r})$$

because the distribution on the right is precisely the one that integrates to $4\pi k$ over any volume containing the origin and to 0 otherwise.

3. Relation to the Laplacian of $1/r$ Since $\mathbf{F} = -k\nabla(1/r)$, taking the divergence gives

$$\nabla \cdot \mathbf{F} = -k \nabla^2 \left(\frac{1}{r} \right).$$

Comparing with the previous boxed result yields the standard identity

$$\nabla^2 \left(\frac{1}{r} \right) = -4\pi \delta^{(3)}(\mathbf{r}).$$

Remark (distributional sense) All statements above about $\nabla \cdot \mathbf{F}$ and $\nabla^2(1/r)$ must be interpreted distributionally: point wise equalities hold for $r \neq 0$, while at the origin the singular behavior is captured by the Dirac delta. Equivalently, for any smooth test function φ with compact support,

$$\int_{\mathbb{R}^3} \nabla^2 \left(\frac{1}{r} \right) \varphi(\mathbf{r}) d^3r = -4\pi \varphi(\mathbf{0}).$$

1.2 The Electric Field

1.3 Boundary Conditions, work and energy, conductors

1.4 Calculating Scalar Potentials I

1.5 Calculating Scalar Potentials II

1.6 Dielectrics

1.7 Electric Displacement

1.8 Energy in dielectrics, Magnetostatics

1.9 Magnetic Vector Potential

1.10 Unit Summary

Unit 2

Magnetostatics